SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE	DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

Program Name: B. Tech	**Assignment Type: Lab**	**Academic Year:**2025-2026
Course Coordinator Name	Dr. Rishabh Mittal	
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CourseCode	23CS002PC304	**Course Title**	AI Assisted Coding
Year/Sem	III/II	**Regulation**	R23
Date and Day of Assignment	**Week9 – Monday**	**BATCH**	B-48
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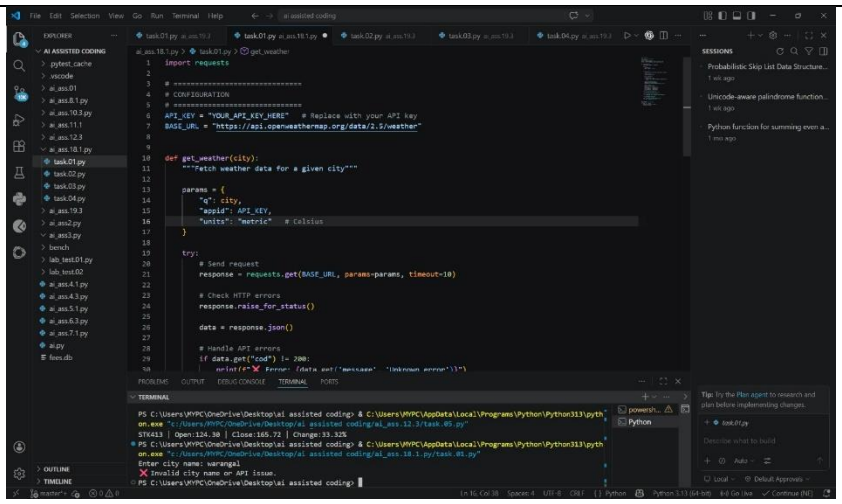
Assignment Number: 18.1(Present assignment number)/**24**(Total number of assignments)

Q.No.	**Question**	***Expected Time to complete***
1	Lab 18 – API Integration: Connecting to External Services with Error Handling	Week9 - Monday

	Lab Objectives: • Learn how to integrate Python programs with external REST APIs. • Understand API request/response handling using requests or similar libraries. • Implement proper error handling for failed API calls (timeouts, invalid responses, rate limits). • Practice extracting and displaying meaningful information from API responses.	
	Task 1 – Weather API Integration Connect to a Weather API to fetch current weather details for a given city. Instructions: • Prompt AI to generate Python code using the requests library. • Handle errors: invalid city name, API key errors, and network failures. • Display temperature, humidity, and weather description in a readable format. **PROMPT :** Explain how to integrate a Weather API in Python to fetch real-time weather data for a given city. Describe how API requests are sent, responses are processed, and how error handling is implemented for invalid inputs and network failures.	



Expected Output:

- A working Python script that fetches and displays weather data with proper error handling.

EXPLANATION : In this task, we connect a Python program to an external Weather API using the `requests` library. The API sends back data in JSON format when we provide a city name and API key.The program extracts useful details such as temperature, humidity, and weather description from the response. Error handling is very important here:

- If the city name is incorrect → API returns an error.
- If API key is invalid → authentication error occurs.
- If network fails → request exception is handled.

This ensures the program runs smoothly without crashing and gives meaningful messages to the user.

Task 2 – Currency Exchange Rates

Integrate a Currency Exchange API to convert INR to USD, EUR, and GBP.

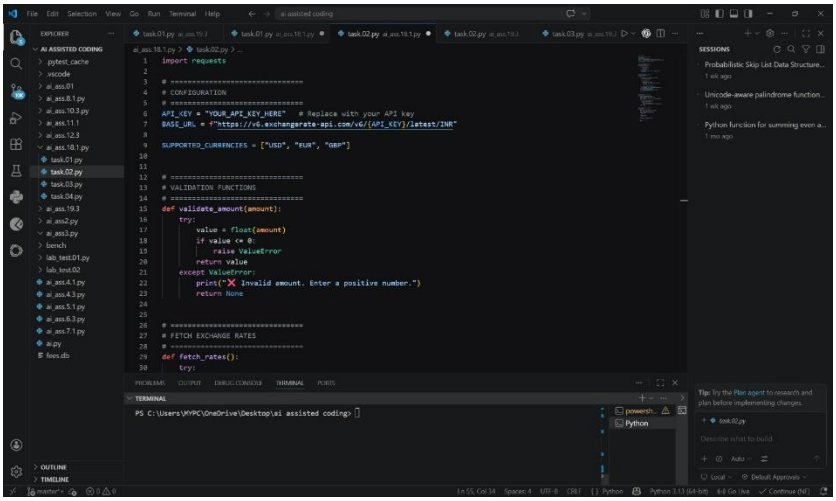
Instructions:

- Use AI to generate API request code.
- Validate user input for amount and currency codes.
- Implement error handling for invalid responses and API

downtime.

- Show conversion results clearly in tabular format.

PROMPT : Describe how to use a Currency Exchange API to convert INR into multiple currencies. Explain input validation, API request handling, and error management for incorrect inputs and API failures.



Expected Output:

- A script that converts INR to multiple currencies with graceful error handling.

EXPLANATION : This task focuses on converting Indian Rupees (INR) into currencies like USD, EUR, and GBP using an external API.

Steps involved:

- Take user input (amount in INR)
- Validate input (must be numeric and positive)
- Send API request to get exchange rates
- Perform conversion using received rates

Error handling includes:

- Invalid input (non-numeric values)
- API downtime or failure
- Incorrect currency codes

The result is displayed clearly in a structured format (like a table), making it easy to understand.

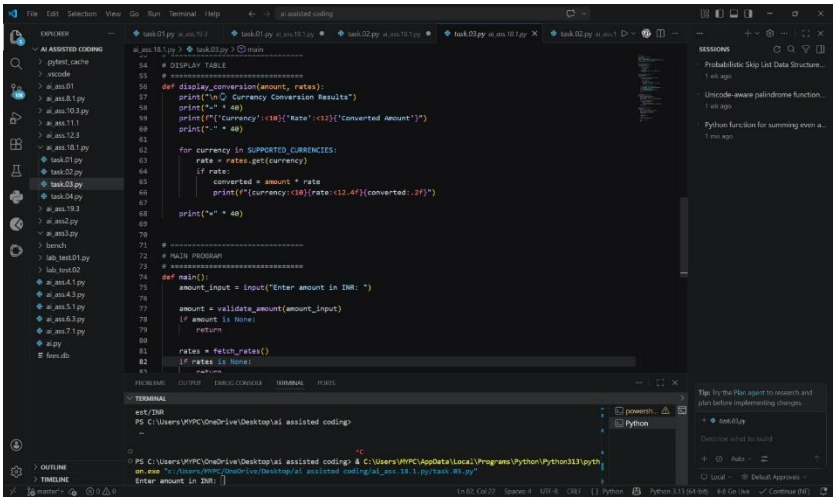
Task 3 – GitHub Repository Info Fetcher

Connect to the GitHub API to fetch details of a repository (e.g., stars, forks, issues).

Instructions:

- Prompt AI to write a Python function to send GET requests to GitHub API.
- Handle rate limit errors, 404 (repo not found), and invalid input.
- Display repository name, description, stars, forks, and open issues.

PROMPT : Explain how to fetch repository details from GitHub API using Python. Describe how GET requests work and how errors like rate limits, invalid repositories, and missing data are handled.



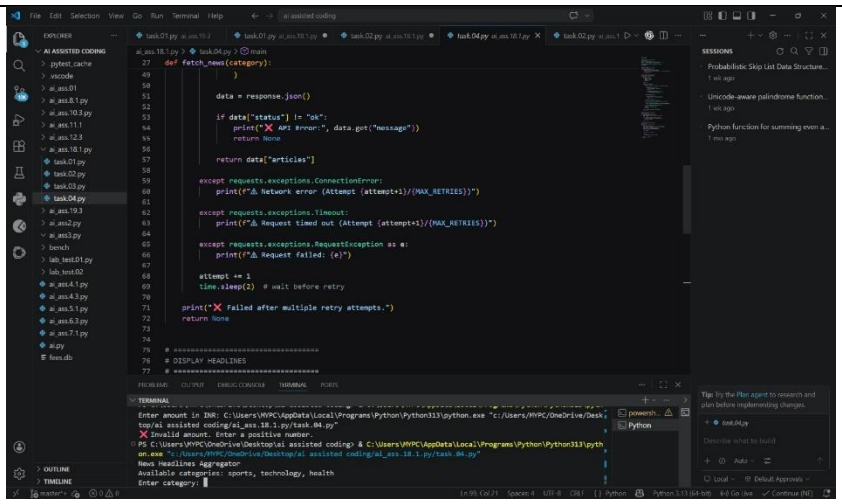
Expected Output:

- A Python program that retrieves and displays GitHub repository information with robust error handling.

EXPLANATION : In this task, we use the GitHub API to retrieve repository information such as:

- Repository name
- Description
- Stars

	• Forks • Open issues The Python program sends a GET request to the GitHub API endpoint. The response is in JSON format, from which required details are extracted. Error handling includes: • 404 Error → Repository not found • Rate limit exceeded → API blocks further requests • Invalid input → wrong repository name This makes the program reliable and user-friendly.	
	Task 4 – Real-Time Application: News Headlines Aggregator Scenario: Build a News Aggregator using a free News API. Requirements: • Fetch top 5 headlines for a given category (sports, technology, health). • Handle errors like invalid category, missing API key, and request failures. • Display headlines in a user-friendly numbered list format. • Include a retry mechanism if the first request fails. **PROMPT :** Explain how to build a news aggregator application using a News API. Discuss fetching headlines, handling errors, and implementing retry mechanisms for failed requests.	



Expected Output:

- A working Python script that retrieves and displays news headlines with proper error handling

EXPLANATION : This task involves creating a real-time application that fetches top news headlines based on categories like sports, technology, or health.

Steps:

- User selects a category
- API request is sent to fetch news
- Top 5 headlines are extracted and displayed

Error handling includes:

- Invalid category input
- Missing or wrong API key
- Network or request failure

A retry mechanism is implemented so that if the request fails initially, the program tries again automatically. This improves reliability.

The output is displayed in a numbered list, making it easy to read.